Chung 2017 breast cancer — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) on sample: per-neighbourhood same-sample fraction 0.75→0.25 (chance 0.09); iLISI 3.8→5.9 of 11 samples. One connected component, 0 detached islands. Samples intermix across a single continuous strip — GO.

Gate 2 — module geography → GO (non-branching)

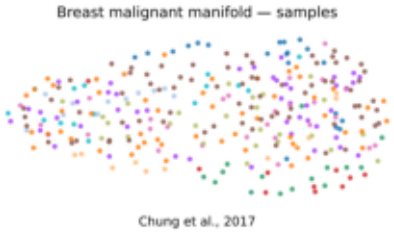
Gast & pSom co-vary (raw r=+0.18) and both oppose Crest (r=−0.65 / −0.54). Along pseudotime Gast/pSom fall (ρ≈−0.95) while Crest rises (ρ=+0.95); NC-high decile at t=0.98 (terminal); joint pSom&Crest enrichment 0.37<1 — inverse terminal gradient, a single (Gast/pSom)→Crest axis. Present but not resolved into two arms — GO.

Gate 3 — cross-repression → NA

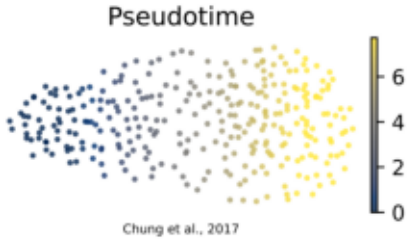
Non-branching topology (curve: 2 tips, 0 forks across N=15/20/30; PPT tree also 0 forks). No bifurcation to test — NA.

STEP FIGURES

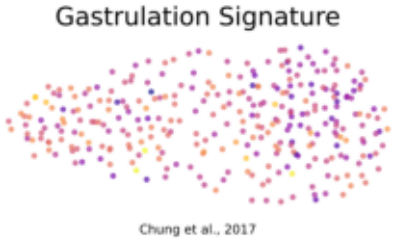
Manifold — samples



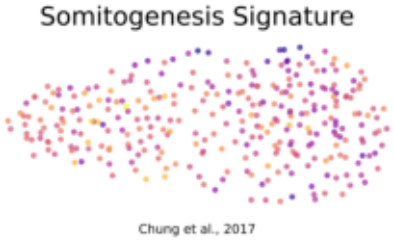
Pseudotime



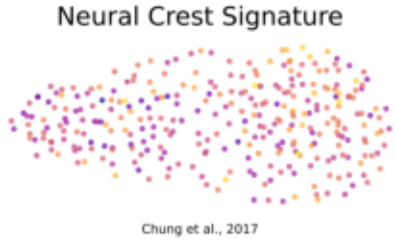
Gastrulation



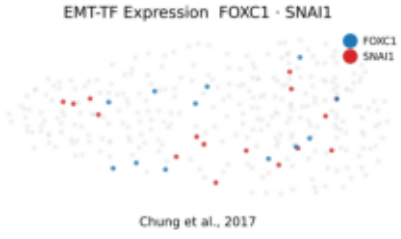
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (Chung et al., 2017), SmartSeq2 — 317 malignant cells (13 samples reported) → 309 analysed (11 samples ≥10 cells). Sparse plate-based data. Non-branching Linear-pSom/Crest.

Verdict.
The embryonic pattern reproduces as a single gastrulation/somitogenesis → neural-crest axis (non-branching); no Gast↔pSom bifurcation, so Gate 3 is NA. Geometry Linear-pSom/Crest — consistent with the linear outcome across most CCCA datasets. EMT-TFs FOXC1 (early/root) → SNAI1 (late/terminus).

Parameter choices

Threshold	≥10 cells/sample (SmartSeq2 scale; anchor floor 100 would empty it) → 11/12, 309 cells	**Embedding**	Harmony θ=10; n_pcs=20, n_neighbors=30, spread=1.3; 401 HVG
Trajectory	scFates curve (non-branching), N=20, λ=0.3, μ=100; fold-free (2.98)	**Root**	Gast/pSom-high, Crest-low tip (Gast/pSom early, Crest late); lower-cycling pole
Geometry	Linear-pSom/Crest; pSom globally strongest (Gast −0.276, pSom +0.124, Crest −0.074)	**EMT-TFs**	FOXC1 (early/root) → SNAI1 (late/terminus); 10-TF panel, VIM excluded